

File 34: GT Intern Workstream — Page 3: AutoCAD Civil 3D Bioengineering Structures Design

Hydrologic Engineering & Geomorphic Design Manual Target School: Georgia Institute of Technology (Georgia Tech) School of Civil & Environmental Engineering Project Area: Upper Savannah Basin & Chattahoochee River Spawning Headwaters Supervising Board: Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)

1. Design Philosophy: Natural channel Re-Meandering

In high-gradient trout streams, standard concrete retaining walls or rock rip-rap dumping are ecologically unacceptable. They destroy aquatic habitat, block macroinvertebrate colonization, and reflect high velocities downstream, causing accelerated erosion elsewhere.

Instead, you will draft **Natural Channel Design (NCD)** and bioengineered structures using native materials: red cedar logs, oak root-wads, and high-strength geogrid soil lifts. These structures stabilize eroding banks, scour deep cold pools, and mimic historic mountain stream features. Your role as a GT engineering intern is to generate precise, AutoCAD Civil 3D plan-sheets, cross-sections, and detail diagrams.

2. Technical Drawings & Dimensions for NCD Structures

You will draft four primary bioengineered structures on our site plans:

A. Rock J-Hook Vane (Flow Deflection & Pool Scour)

- **Description:** A rock structure built along the outer bank of a curve. The "vane" portion slopes upstream, directing fast currents away from the bank, while the downstream "hook" scours a deep central pool.
- **AutoCAD Parameters:**
- **Vane Angle:** Set between 20° and 30° relative to the upstream bank tangent.
- **Slope:** The rock arm must slope down from the bankfull elevation at the bank to the stream bed at $1/3$ the width of the channel.
- **Hook Rocks:** Position the J-hook rocks in the center of the channel, submerged below the baseflow water surface. Leave gaps of $10\text{--}15\text{ cm}$ between rocks to allow trout passage and scour halos.

B. Double Log Cross-Vane (Grade Control & Pool Scour)

- **Description:** A two-arm log structure crossing the channel, sloping upstream to create a central plunge pool and prevent bed incision (erosion of the stream bed).
- **AutoCAD Parameters:**

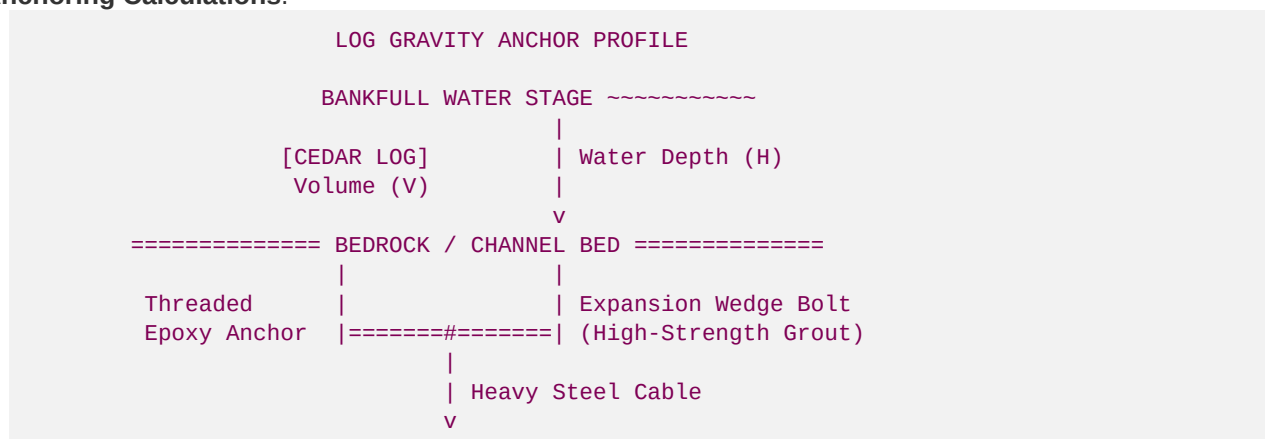
- **Log Material:** Select fresh red cedar or white oak logs ($30\text{--}45\text{ cm}$ diameter).
- **Throat Width:** The central "weir" gap must span exactly $\frac{1}{3}$ the bankfull width of the channel.
- **Log Arm Slope:** Slope the log arms upward at $2\text{--}7\%$ from the stream bed to the bankfull bank line.

C. Cedar Toe-Wood (Bank Stabilization & Woody Habitat)

- **Description:** Layers of overlapping cedar logs and root-wads placed at the toe of a regraded bank curve, covered with soil lifts.
- **AutoCAD Parameters:**
- **Placement:** Place the cedar logs below the baseflow water line to prevent decay from air exposure.
- **Root-wad Projection:** Extend the fibrous cedar root-wads outward into the stream channel at a 45° angle facing upstream to trap fine gravels and provide cover for large brown trout.

3. Structural Anchoring & Geotech Calculations

To ensure these cedar structures do not wash away during major floods, you must execute the **Log Gravity Anchoring Calculations**.



The buoyant upward force (F_b) acting on a submerged log must be completely countered by the downward gravitational force of the soil/rock overburden and mechanical anchor anchors.

Anchor Safety Factor Equation:

$$FS_{\text{anchor}} = \frac{W_{\text{overburden}} + W_{\text{log_dry}}}{F_{\text{buoyant}}} \geq 1.5$$

Where:

- $F_{\text{buoyant}} = \gamma_{\text{water}} \cdot V_{\text{log}}$, where $V_{\text{log}} = \pi \cdot R^2 \cdot L$ and $\gamma_{\text{water}} = 62.4\text{ lb/ft}^3$.
- $W_{\text{log_dry}} = \gamma_{\text{cedar}} \cdot V_{\text{log}}$, where dry red cedar weight $\gamma_{\text{cedar}} \approx 23.0\text{ lb/ft}^3$.
- $W_{\text{overburden}} = \text{weight of rocks/soil placed on top of the log keys in the stream bank}$.

Worked Example:

Calculate the buoyant force and required anchoring weight for a cedar log of radius $R = 0.75\text{ feet}$ and length $L = 12.0\text{ feet}$:

1. Calculate Log Volume ($V_{\log}$):

$$V_{\log} = \pi \cdot (0.75)^2 \cdot 12 = 21.21 \text{ ft}^3$$

2. Calculate Buoyant Force (F_{buoyant}):

$$F_{\text{buoyant}} = 62.4 \cdot 21.21 = 1,323.5 \text{ lbs of upward force}$$

3. Calculate Log Dry Weight ($W_{\log_dry}$):

$$W_{\log_dry} = 23.0 \cdot 21.21 = 487.8 \text{ lbs of downward dry weight}$$

4. Calculate Required Overburden/Mechanical Anchor Weight (W_{req}) for $FS = 1.5$:

$$1.5 = \frac{W_{req} + 487.8}{1,323.5} \text{ implies } W_{req} = (1.5 \cdot 1,323.5) - 487.8 = 1,497.45 \text{ lbs}$$

Your design must specify a minimum of **1,500 lbs of mechanical cable-anchors** (epoxied into bedrock or secured using concrete/rock deadman anchors in the bank) to satisfy the $FS_{\text{anchor}} \geq 1.5$ safety margin.

4. Geotechnically-Reinforced Soil Lift (GRSL) Details

To rebuild vertical, stable banks along outer bends, you will detail **Geotechnically-Reinforced Soil Lifts (GRSL)**:

- **Core Fabric:** Specify high-durability, double-walled **coir fiber geotextile wrap (Type 700 or 900)**.
- **Filling:** Pack the inside of the fabric wraps with bank-run compacted gravel (15 cm base) and rich organic soil (15 cm top) to promote native root growth.
- **Compaction:** Specify 95% Standard Proctor Density compaction for the soil core.
- **Interlocking:** Wrap the coir fabric around the soil layers in overlapping envelopes, stepping the lifts back at a 1:0.5 (Horizontal to Vertical) slope. Interleave native willow stakes and tag alder branches between the lifts to bind the system mechanically.